

Euler's Pentagonal Number Theorem

Formalization via Franklin's involution & the Jacobi triple product

JC, PM, MV

May 24, 2026

The theorem

Euler's Pentagonal Number Theorem

$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2} = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{\frac{k(3k-1)}{2}} + x^{\frac{k(3k+1)}{2}} \right).$$

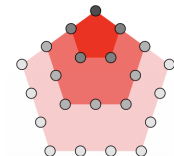
The theorem

Euler's Pentagonal Number Theorem

$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2} = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{\frac{k(3k-1)}{2}} + x^{\frac{k(3k+1)}{2}} \right).$$

Combinatorial form. Let $p_e(n)$ (resp. $p_o(n)$) count partitions of n into an *even* (resp. *odd*) number of *distinct* parts. Then

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & n = \frac{3k^2 \pm k}{2} \text{ (pentagonal)}, \\ 0 & \text{otherwise.} \end{cases}$$



k	-3	-2	-1	0	1	2	3
$\frac{3k^2-k}{2}$	15	7	2	0	1	5	12

Generalized pentagonal numbers $\frac{3k^2-k}{2}$ ($k \in \mathbb{Z}$): 0, 1, 2, 5, 7, 12, 15, ...

Base, slope, and Franklin's involution

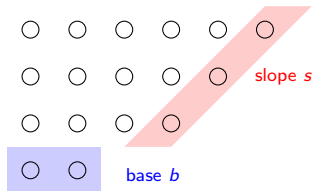


Diagram of $17 = 6 + 5 + 4 + 2$.

$b = \min(S)$, $m = \max(S)$,
 $s = \text{length of the top consecutive run}$,
 $D = \{m - s + 1, \dots, m\}$ the slope set.

A partition lies in exactly one class:

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$$

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}$$

Proof sketch (Franklin's involution)

Define $F = \alpha$ on $\mathcal{P}_\alpha$, β on $\mathcal{P}_\beta$, and the identity on $\mathcal{P}_{\text{special}}$.

- ① $\alpha : \mathcal{P}_\alpha \rightarrow \mathcal{P}_\beta$ and $\beta : \mathcal{P}_\beta \rightarrow \mathcal{P}_\alpha$ are mutually inverse $\Rightarrow F$ is an **involution** with $\beta \circ \alpha = \text{id}$, $\alpha \circ \beta = \text{id}$.
- ② F **preserves n** and **flips the parity** of the number of parts on $\mathcal{P}_\alpha \cup \mathcal{P}_\beta$.
- ③ Hence α, β pair off even with odd partitions: they cancel in $p_e - p_o$.
- ④ Only the **fixed points** $\mathcal{P}_{\text{special}}(n)$ survive: empty unless n is pentagonal, where it is a single staircase $\{k, \dots, 2k-1\}$ or $\{k+1, \dots, 2k\}$ of parity $(-1)^k$.

$$p_e(n) - p_o(n) = \sum_{S \in \mathcal{P}_{\text{special}}(n)} (-1)^{|S|} = \begin{cases} (-1)^k & n = \frac{3k^2 \pm k}{2} \\ 0 & \text{else.} \end{cases}$$

Lean formalization (Aristotle/) — $\approx 1,036$ lines, sorry-free

Component (file)	Lemmas	Lines	Effort
Definitions (Defs)	$\mathcal{P}_\alpha, \mathcal{P}_\beta, \mathcal{P}_{\text{spec}}, \text{base/slope}$	90	Low
Helpers (Helpers)	$\text{run/slope}, b \leq m, m \geq 2b$	322	High – manual induction
Disjoint + trichotomy	Lem. 11–12	~ 26	Low – grind/simp
Special = pentagonal	Lem. 13–14	~ 230	Med. – arithmetic
Franklin involution	Lem. 16–19	~ 120	High – very manual
Bijection & counting	Lem. 20–24	~ 135	Med. – Finset.card

Easy: definitions, disjointness/trichotomy (one grind each), final assembly.

Hard / very manual: reasoning about *slope* (a recursive top-run) and *base/max* under $\cup/\setminus$; the four involution lemmas $\alpha \in \mathcal{P}_\beta, \beta \in \mathcal{P}_\alpha, \beta \circ \alpha = \text{id}, \alpha \circ \beta = \text{id}$.

Library-assisted: cardinality/bijection step rides on mathlib's Finset.

Three formalizations of the *same* bijective proof

All three prove Euler's PNT via Franklin's involution; they differ in how much structure is spelled out, the size of the input prompt, and whether the proof

	Aristotle	SlidesInput	OnlyWordsInput
closes.			
Total lines	1,350	728	460
Files	5	3	3
Status	sorry-free	sorry-free	2 sorrys

- **Aristotle (1,350)**: explicit *three-class* split $\mathcal{P} = \mathcal{P}_\alpha \sqcup \mathcal{P}_\beta \sqcup \mathcal{P}_{\text{spec}}$ with *two separate* maps α, β proven mutually inverse, plus a `FormalPowerSeries` layer linking $p_e - p_o$ to $\prod(1 - X^i)$. Most verbose, fully self-contained.
- **SlidesInput (728)**: *one* map `franklinMap` (ops A/B via a staircase length `stairLen`); non-fixed points cancel, fixed points characterized as the two staircases $\{k..2k-1\}$, $\{k+1..2k\}$. Same content, tighter packaging.
- **OnlyWordsInput (460)**: leanest — rides `mathlib`'s `Finset.sum_involution` and a `topRun` helper. Closes the cancellation argument but leaves the involutivity and fixed-point lemmas as `sorry`.

Same mathematics, $\sim 3\times$ length spread: more guidance / more library reuse $\Rightarrow$ shorter code, but the shortest run did not fully close.

Jacobi triple product (plain form)

Jacobi triple product

$$\prod_{m=1}^{\infty} (1 - q^{2m}) (1 + q^{2m-1} w^2) (1 + q^{2m-1} w^{-2}) = \sum_{k=-\infty}^{\infty} q^{k^2} w^{2k}.$$

Equivalently, with $z = -w^2$,

$$(q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty} = \sum_{k \in \mathbb{Z}} z^k q^{k(k-1)/2}.$$

A single bilateral identity that specializes to many classical q -series results.

q -Pochhammer symbols and the q -binomial theorem

Finite / infinite q -Pochhammer symbol:

$$(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k), \quad (a; q)_\infty = \prod_{k=0}^{\infty} (1 - aq^k).$$

Gaussian binomial coefficient: $\binom{n}{k}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}.$

q -binomial theorem (finite, then $n \rightarrow \infty$)

$$\prod_{k=0}^{n-1} (1 + zq^k) = \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k, \quad \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_\infty}{(z; q)_\infty}.$$

The two Euler identities follow by $a = 0$ and $a \rightarrow \infty$; the Jacobi triple product is then a short corollary. (*Lean*: `qBinom_finite_thm`, `qBinom_infinite_thm` — *high up-front cost, low marginal.*)

Euler's PNT from the Jacobi triple product

Start from the Jacobi triple product

$$\prod_{m \geq 1} (1 - q^{2m})(1 - q^{2m-1}z)(1 - q^{2m-1}z^{-1}) = \sum_{k \in \mathbb{Z}} (-1)^k z^k q^{k^2}.$$

Euler's PNT from the Jacobi triple product

Start from the Jacobi triple product

$$\prod_{m \geq 1} (1 - q^{2m})(1 - q^{2m-1}z)(1 - q^{2m-1}z^{-1}) = \sum_{k \in \mathbb{Z}} (-1)^k z^k q^{k^2}.$$

Specialize $q \mapsto x^{3/2}$, $z \mapsto x^{1/2}$:

$$\underbrace{\prod_{m \geq 1} (1 - x^{3m})(1 - x^{3m-1})(1 - x^{3m-2})}_{= \prod_{n \geq 1} (1 - x^n)} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2+k)/2}.$$

Euler's PNT from the Jacobi triple product

Start from the Jacobi triple product

$$\prod_{m \geq 1} (1 - q^{2m})(1 - q^{2m-1}z)(1 - q^{2m-1}z^{-1}) = \sum_{k \in \mathbb{Z}} (-1)^k z^k q^{k^2}.$$

Specialize $q \mapsto x^{3/2}$, $z \mapsto x^{1/2}$:

$$\underbrace{\prod_{m \geq 1} (1 - x^{3m})(1 - x^{3m-1})(1 - x^{3m-2})}_{= \prod_{n \geq 1} (1 - x^n)} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2+k)/2}.$$

The three residue classes mod 3 reassemble the full product $\prod_{n \geq 1} (1 - x^n)$, and $(3k^2 + k)/2$ ranges over the pentagonal numbers. Hence

$$\prod_{n \geq 1} (1 - x^n) = \sum_{k \in \mathbb{Z}} (-1)^k x^{k(3k-1)/2}$$

— exactly Euler's Pentagonal Number Theorem, now an analytic corollary rather than a bijective one.

Formalizing the q -series route: where Aristotle struggled

Unlike the Franklin-involution PNT (**one** Aristotle run each), the Jacobi triple product took **~ 6 documented runs** to close — two of which produced *no code*, only cost/feasibility analyses.

- **Two planned routes abandoned.** (i) Complex analysis / Liouville: needs holomorphicity of infinite products and a Liouville theorem on $\mathbb{C}^*$ — **not in mathlib** (~ 900 – 1500 lines of new infrastructure). (ii) Hermite-style coefficient recurrence ($A_{n,k}$ bookkeeping): ~ 1500 – 2000 lines. Both judged too costly.
- **The crux: annulus $\rightarrow$ punctured disc.** The Cauchy-product proof first only worked on $\|q\| < \|z\| < 1$; the naive extension hit **boundary cases $\|z\| = \|q\|^n$** and stalled at one stubborn sorry for a full run.
- **Solution (Aristotle).** Prove the 2nd Euler identity for *all* z (not just $\|z\| < 1$) via the series recurrence $S(z) = (1 + z) S(qz)$ — this **removes the annulus restriction entirely**, and the product argument runs directly on $0 < \|z\| < 1$.
- **Infrastructure-heavy core.** Cauchy product over $\mathbb{N} \times \mathbb{N}$ (Fubini + diagonal split), each diagonal evaluated by the Durfee-rectangle identity $S_k = 1/(q; q)_\infty$ — proved by an *analytic contraction recurrence*, not the textbook bijection.

Trajectory: 3 sorrys $\rightarrow 1 \rightarrow 0$ across runs; final $\sim 1,700$ -line sorry-free library — but only after iteration and planning the bijective proof never needed.

LLM usage

Model	Output	Input	Cache write	Cache read
Opus 4.7	4.1 M	61 K	25 M	456 M
Sonnet 4.6	1.02 M	0.6 K	2.76 M	21.4 M
Total (interactive, est.)	5.1 M	62 K	28 M	477 M
Aristotle (est.)	$\approx 50\text{--}130\text{ M total} \text{ — } \sim 8\text{--}12 \text{ runs total}$			

- **Interactive cost @ API rates** (standard Opus/Sonnet tiers, *not* the Max plan; recorded + this session, est.): $\approx \$1,500$ (\$1,465 Opus + \$32 Sonnet; output \$310, cache-read \$685, cache-write \$470), $\sim 15\text{ kWh}$ compute / $\sim 19\text{ kWh}$ with datacenter overhead — $\approx$ driving an EV $\sim 110\text{ km}$, or a fridge for ~ 2 weeks.
- **Aristotle (est. — no token logs; from run counts):** OnlyWordsInput 1 run / 460 ln / 2 sorrrys; SlidesInput 1 run / 728 ln / closed; qSeries+JTP **6 documented runs** (2 spent only on feasibility/cost analysis) / 1.7 k ln / closed. Total $\sim 8\text{--}12$ runs, $\approx 50\text{--}130\text{ M tok}$, $\approx \$200\text{--}800$, $\approx 4\text{--}20\text{ kWh}$.
- **Punchline:** the *bijective* PNT closed in *one* Aristotle run each (cheap, one-shot); the *analytic* JTP route needed ~ 6 iterative runs — two spent only *planning* — yet the whole Aristotle effort ($\approx \$200\text{--}800$) still came in *below* the interactive exploration around it ($\approx \$1,500$). Easy proofs are one-shot; hard ones iterate — but generating proofs stays cheaper than talking about them.